

STUDENT TRACK

14TH INTERNATIONAL CONFERENCE ON CONTEMPORARY ISSUES IN MANAGEMENT

20th and 21st February 2026

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*Title of the Paper: India's Growth Path in BRICS: Evidence from
Convergence Analysis of Key Economic Indicators*

CMT Paper ID: 97

Introduction

- BRICS accounts for over 45% of the world's population, 37% of the global GDP (PPP) and 23% of global trade
- Assessing internal cohesion and long run sustainability is critical, especially since it aims to institutionalise economic cooperation through NDB and CRA
- Analysing India's position in the bloc is necessary due to its emerging significance as a military, economic and political power
- Study uses convergence tests and median comparison on GDP per capita, GDP growth rate, FDI Inflows and Trade Openness

Review of Literature

- Convergence can be tested in several ways: β -convergence, σ convergence, stochastic convergence and club convergence (Bernard & Durlauf, 1995; Phillips & Sul, 2007; Islam, 2003, Iranmanesh, 2025).
- Prior studies highlight that convergence results within BRICS are mixed and sensitive to methodology, with some finding only partial or conditional convergence, and many emphasizing dependence on structural variables and growth drivers (Banday et al., 2021; Lohani, 2021; Das & Nayak, 2023; Phiri, 2018).

Review of Literature

- Members display differences in economic size, structure and policy complicates direct comparison, rendering group averages potentially misleading (Armijo, L. E., 2007)
- GDP per capita is our main variable as it has been found to be the most adequate measure of economic welfare (Phiri, 2008), while GDP growth rate looks at the economy as a whole
- Trade openness and FDI are found to be the key drivers of economic growth in the globalized world, justifying their usage in the study (Banday et al., 2021)

Research Gap/Research Problem Statement

- Existing literature overwhelmingly analyses the bloc as a whole
- Study will help to evaluate if the bloc economy indeed acts as a whole
- India specific study does not exist, as far as the researcher's knowledge is concerned
- Addressing this gap is necessary to aid economists and policy makers in assessing internal balance and formulating bloc-level policies

Research Objectives & Research Hypotheses

- To access the relative position of India within BRICS based on median comparison and its trajectory over time
- To evaluate economic homogeneity in BRICS and determine if it functions as a unit
- To test for σ -convergence and evaluate if dispersion of economic indicators is reducing or widening
- To test for β -convergence and evaluate catch-up dynamics in the bloc.
- To investigate the existence of club convergence and evaluate if the club can fracture into rich and poor clubs

Research Methodology

- Temporal scope: 2006 – 2024 (19 years)
- Geographical scope: India, Russia, China, Brazil and South Africa (the five member nations)
- BRICS partner nations excluded to maintain longitudinal consistency
- Variables: GDP per capita, GDP growth rate, FDI Inflows as a percentage of GDP, Trade Openness as a percentage of GDP
- Data Source: World Bank Group's World Development Indicators database to ensure cross comparability and reliability
- Statistical Tool: R Programming

Research Methodology

- Median Comparison: India is ranked among the bloc members based on its positioning relative to the annual median to identify its trajectory over the years
- σ -convergence: Cross-sectional dispersion of economic indicators across BRICS members is tested to assess if economies are becoming similar over time
- β -convergence: A panel regression is employed to test if poor countries are growing faster than richer ones, indicating catch-up dynamics
- Club convergence: Phillips and Sul's (2007) log-t regression test is employed to evaluate whether the economies share a common long-run path of evolution

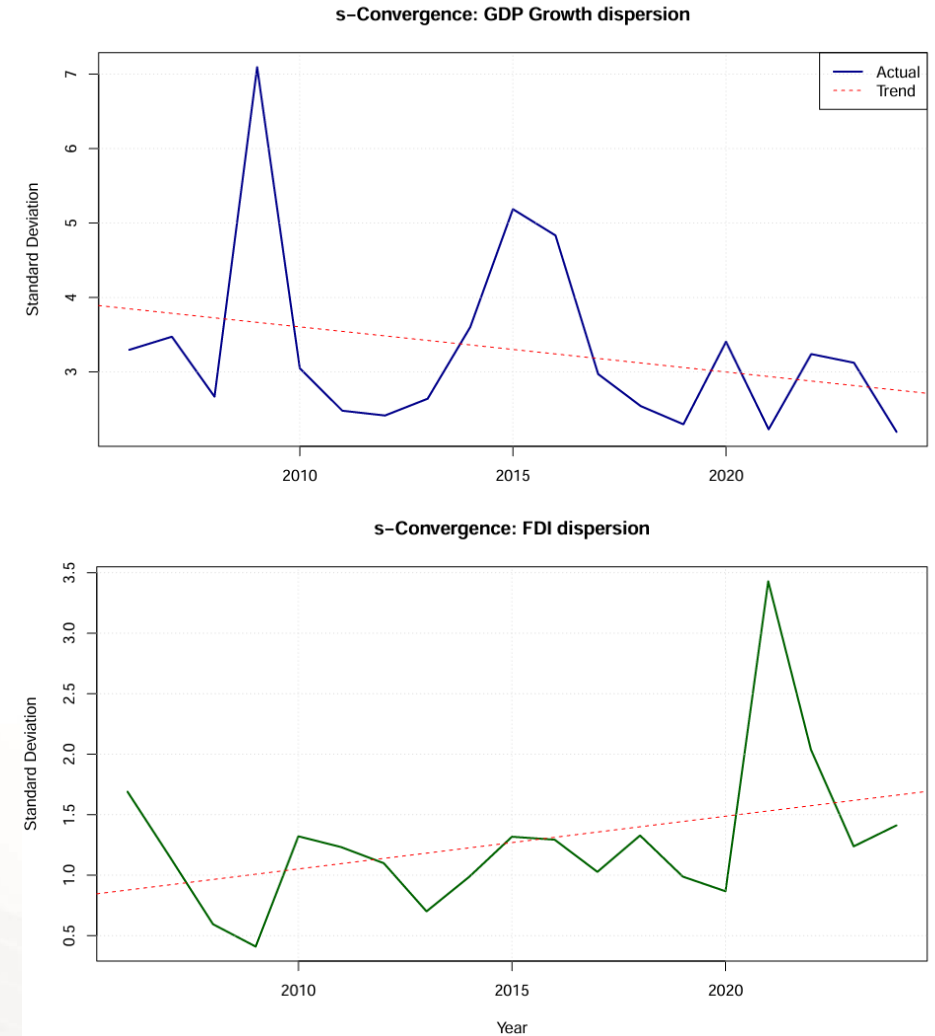
Analysis and Interpretations

Median Comparison:

Variable	Median % in 2006	Median % in 2024	Additional Remark
GDP Per Capita	30.6%	50%	Ranks last every year, Poorest country in the bloc
GDP Growth Rate	100%	149%	Top 3 every year apart from 2008 and 2020, #1 in 6 out of the remaining 17 years
FDI Inflows	100%	119%	Attracted investment even in unfavorable conditions
Trade Openness	85%	113%	Significantly opened economy, corroborates with Kumar et al. (2023)

Analysis and Interpretations

- σ -convergence:
 - Per Capita GDP: Dispersion decreased, 27.7% reduction in inequality
 - GDP Growth Rate: Dispersion decreased, 33.4% reduction in inequality.
 - Trade Openness: Dispersion decreased, 25.1% reduction in inequality.
 - FDI Inflows: Dispersion decreased until 2009, the trend reversed to increase inequality by 245%.
- The variables exhibited divergence during global shocks but with lesser magnitude over time, correcting soon after the shock



Analysis and Interpretations

- β -convergence:
 - Unconditional: Catchup speed is 2.5%; Result is insignificant, better than just luck due to small sample bias
 - Conditional (Fixed Effects model): Catchup speed is 4.8%; Highly significant with more than 99% confidence

term	estimate	std.error	statistic	p.value
(Intercept)	0.263048	0.141318	1.861391	0.15961
log_initial	-0.02503	0.015102	-1.6575	0.195999

term	estimate	std_error	t_value	p_value
			-	
log_gdp_pc_lag	-0.04802	0.014696	3.26726	0.001574

- Club Convergence:
 - Converging to a single equilibrium for trade openness and GDP growth rate
 - GDP per capita: Russia and China in Club 1
 - FDI Inflows: India and Brazil in Club 1

Findings and Implications

- 27.7% reduction in gap between the richest and the poorest country
- Existence of clubs show internal cohesion is fractured in parts
- India is identified as the strongest driver of convergence, capable of preventing bloc fragmentation
- It underscores the need for a bloc-wide economic policy understanding and bridge gaps if BRICS wants to counter existing institutions

Findings and Implications

- Structural differences play a major role in economic development
- India must utilize its FDI Inflows to open its trading model to align closer with the developed economies
- Small sample bias is shown, convergence is directionally correct however the magnitude may be different slightly

Conclusion

- The research provides necessary insights to policy makers and economists to deepen their understanding of India's role in BRICS
- Provides insights into economic bottlenecks, formation of BRICS currency and development of the New Development Bank
- Future scope of research includes addition of BRICS partner countries to remove small sample bias as well as to evaluate their role in the bloc, considering structural differences post-pandemic recovery phase during convergence analysis

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